

Quiz 1

Section: _____

Student ID: _____

Total Marks: 15

A national research fellowship board uses a machine learning model to shortlist candidates for funding. Recently, concerns have been raised that candidates with very similar academic and research profiles may be receiving different outcomes, possibly indicating unintended bias.

To audit this issue, the board decides to divide a pool of candidates into two subsets (Batch 0 and Batch 1). The goal is to separate highly similar candidates into different batches so that the model's decisions on similar profiles can be compared.

Let the candidates be:

$$\{C_1, C_2, C_3, C_4, C_5, C_6, C_7, C_8\}$$

The similarity score between candidates (computed from GPA, publications, research experience, and recommendation strength) is given below. Higher values indicate greater similarity.

	C_1	C_2	C_3	C_4	C_5	C_6	C_7	C_8
C_1	0	9	4	7	3	8	5	6
C_2	9	0	6	2	7	4	8	3
C_3	4	6	0	8	5	3	6	7
C_4	7	2	8	0	9	4	5	6
C_5	3	7	5	9	0	6	4	8
C_6	8	4	3	4	6	0	7	5
C_7	5	8	6	5	4	7	0	9
C_8	6	3	7	6	8	5	9	0

The task is to partition the candidates into two subsets such that:

Answer the following:

- (a) Propose a suitable encoding scheme to represent a candidate partition as a chromosome in a Genetic Algorithm. Clearly explain how a solution is represented.

[4 marks]

- (b) Define a mathematical fitness function to evaluate the quality of a partition. Clearly explain how it satisfies the objective of maximizing cross-subset similarity and minimizing within-subset similarity.

[5 marks]

- (c) Demonstrate (with an example) how single-point crossover would operate between two randomly generated parent chromosomes.

[3 marks]

- (d) If the population size is set to 2, discuss one major problem that may arise during evolution and explain why it affects solution quality.

[3 marks]